Corrective Action Report — CAR-2025-003

Field	Value
Document ID	CAR-2025-003
Issue Date	2025-11-21
Author	S. Whitford, Reliability & Compliance Auditor; reviewed by D. Pham, Compliance Lead
CAR Status	**Open**
Subject	Cooling System Maintenance Procedure CSM-7 Contains Ambiguous Quarterly-vs-Annual Oil-Spectrometer Frequency; Differential Vendor Interpretation Implicated in Fall Overheating Sequence
Affected Region / Site / Asset	NW-PowerPool / NW-PowerPool-WIND-003, NW-PowerPool-WIND-024 / WIN-10085, WIN-10131, TRA-10561, WIN-10011
Severity	High
Owner	K. Driscoll, Manager — Maintenance Engineering Standards
Original Target Close Date	2026-02-28
Distribution	Maintenance Engineering Standards Committee; CPS Vendor Manager; Apex Vendor Manager; NW-PowerPool Plant Manager; Engineering Documents Control

1. Finding

Cooling System Maintenance procedure **CSM-7** ("Gearbox & Coolant Loop Maintenance — Wind Turbine"), revision dated 2022-11-04, prescribes oil-spectrometer sampling under §4.2 with the text:

> "Oil sample shall be drawn and submitted for spectrometer analysis on a scheduled cadence consistent with manufacturer guidance; at minimum, annually, or quarterly for assets identified by Engineering as elevated-risk."

This wording produces inconsistent execution between Cascadia's two principal field-service vendors:

- **Cascadia Power Services Inc. (CPS)** interprets §4.2 as **quarterly** for all NW-PowerPool wind turbines (CPS internally classifies the entire NW-PowerPool wind fleet as elevated-risk based on age and load profile).
- **Apex Emergency Grid Solutions** interprets §4.2 as **annual** unless an Engineering memo is presented to the field crew at the time of service designating the asset as elevated-risk. No such memo exists for most NW-PowerPool turbines.

The gap is not cosmetic. The quarterly interpretation catches early bearing wear via Fe and Cu particle counts approximately 6–9 months before catastrophic vibration onset; the annual interpretation does not. The Fall Overheating Sequence (October–November 2025) appears to be the direct downstream consequence of work performed at NW-PowerPool-WIND-003 and -024 by Apex during the August Period A recovery window — during which oil-spectrometer steps were not executed.

2. Evidence

2.1 Service-record comparison

Audit of work-order ledger 2024-07-01 through 2025-08-31:

Asset	Last Spectrometer Sample (CPS)	Last Spectrometer Sample (Apex visit)	Gap
WIN-10085 (Nordex, NW-PowerPool-WIND-003)	2024-04-12 (CPS)	2025-06-18 (Apex, EMR-2025-003) — NOT performed	14 months
WIN-10131 (Siemens Gamesa, NW-PowerPool-WIND-024)	2024-02-08 (CPS)	2025-05-24, 2025-08-29 (Apex) — NOT performed either visit	21 months by Period B onset
TRA-10561 (Schneider, NW-PowerPool-WIND-003)	2024-09-05 (CPS) — transformer oil, related sample	2025-06-04 (Apex) — NOT performed	15 months by 2025-12-10 failure
WIN-10011 (Siemens Gamesa, NW-PowerPool-WIND-024)	2024-03-22 (CPS)	2025-07-08 (Apex, post-OUT-2025-0320) — NOT performed	20 months

2.2 Linked Period B outages

- OUT-2025-0476 (2025-10-07, WIN-10085, overheating, 226 h, 3,740.6 MWh lost) — post-failure oil analysis showed Fe particle count of 174 ppm (CPS internal alert level: 80 ppm; OEM alert: 120 ppm). Particle count trajectory inferable from quarterly cadence would have triggered intervention in approximately July 2025.
- OUT-2025-1407 (2025-12-10, TRA-10561, overheating, 215 h, 3,558.6 MWh lost) — post-failure dissolved-gas analysis (DGA) abnormal across acetylene, hydrogen, and methane.
- OUT-2025-0319 (2025-12-02, WIN-10011, efficiency_decline, 111 h, 2,706.6 MWh lost) — same root vendor-procedure ambiguity.

2.3 Cost differential

Quarterly spectrometer cadence at $1,850 per sample (CPS catalog) on the 26 NW-PowerPool wind assets is $192,400 annually. The Fall Overheating Sequence direct cost (Apex emergency invoices Period B) was **$3.7M** plus an estimated $1.1M lost generation. Three of the four major Period B outages above are direct candidates for early detection under quarterly cadence.

3. Root Cause Hypothesis

- **Primary:** CSM-7 §4.2 was last revised in 2022 and does not name a default vendor interpretation. It also does not specify the document path by which Engineering communicates "elevated-risk" designation to the field crew.
- **Contributing:** The Engineering elevated-risk list is maintained as an internal SharePoint table updated quarterly; Apex field crews do not have read access to that SharePoint instance. CPS field crews receive the list as a printed addendum to their dispatch packet.
- **Contributing:** No closeout audit exists to verify that the spectrometer step in CSM-7 was actually performed; the digital work-order tick-box accepts a vendor self-attestation without sample submittal verification.

4. Conflict With Existing Guidance

- CSM-7 §4.2 (Cascadia procedure, 2022 revision) — ambiguous wording.
- **VTM-VESTAS-2025 §6.4** prescribes quarterly spectrometer sampling for all assets >7 years in service. This is the rigor model and supports CPS's interpretation.
- **VTM-SIEMENS-GAMESA-4.2 §5.7** prescribes annual sampling, "supplemented by condition-based sampling when nacelle accelerometers exceed Class III ISO 10816-21 levels." This supports Apex's interpretation but is incompatible with the

deferred condition-monitoring environment at NW-PowerPool (see CAR-2025-001 §2.1: WO-2025-00472 and WO-2025-00473 — never executed).

- **VTM-GE-2023** is silent on cadence (the post-2023-firmware refresh manual has not been delivered — see CAR-2025-007).

5. Recommended Corrective Action

1. Reissue **CSM-7 Revision 3** with §4.2 replaced by unambiguous text: oil-spectrometer sampling **quarterly** for all wind turbines >7 years in service, **quarterly** for all transformers in NW-PowerPool ambient envelope (>32 °C summer average), **annually** for assets <7 years. Remove the "Engineering-designation" trigger.
2. Modify CMMS to require sample-submittal lab receipt as a closure artifact for any work order containing a spectrometer step. Auto-reject closure without lab receipt.
3. Issue field memo FM-2025-031 to Apex Vendor Manager establishing the revised cadence as a contractual minimum under EPA-2024-11.
4. Direct CPS to backfill missed spectrometer samples on the 8 NW-PowerPool wind assets identified in §2.1 within 30 days.
5. Establish a quarterly Reliability & Compliance audit of CSM-7 step compliance based on lab submittal records.

6. Action Owner & Timeline

Sub-action	Owner	Target Date
CSM-7 Rev. 3 issued	K. Driscoll	2026-01-15
CMMS lab-receipt requirement	R. Vasquez	2026-02-28
Field memo FM-2025-031 to Apex	M. Beasley (Vendor Mgr)	2026-01-31
Backfill samples on 8 assets	CPS scheduled work	2026-01-31
Quarterly audit process	Reliability & Compliance	2026-03-31

7. Implementation Status

Open. Draft of CSM-7 Rev. 3 circulated 2025-12-08; review comments pending from Engineering Standards Committee. CMMS modification request submitted but unscheduled. Apex Vendor Manager has acknowledged the field memo will be sent but raised that contractual modification under EPA-2024-11 may require renegotiation (cross-reference CAR-2025-004).

Cross-references: CAR-2025-001 (deferred WO-2025-00472, -00473 — direct spectrometer-step deferrals); CAR-2025-004 (Apex contract structure that allows interpretation discretion); CAR-2025-006 (technician fatigue during Period A may have contributed to additional shortcuts of CSM-7 steps); CAR-2025-007 (GE manual silence on cadence); RCA-2025-003 (WIN-10085 root cause); RCA-2025-004 (TRA-10561 December failure); OIR-2025-007 (Period B outage cluster investigation); VTM-VESTAS-2025; VTM-SIEMENS-GAMESA-4.2.

End of report.